

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green. They are positioned diagonally, with the blue one partially covering the green one.

Arcade Cabinet

Brady Garrison, Mechanical Engineering and
Computer Science, Class of 2026



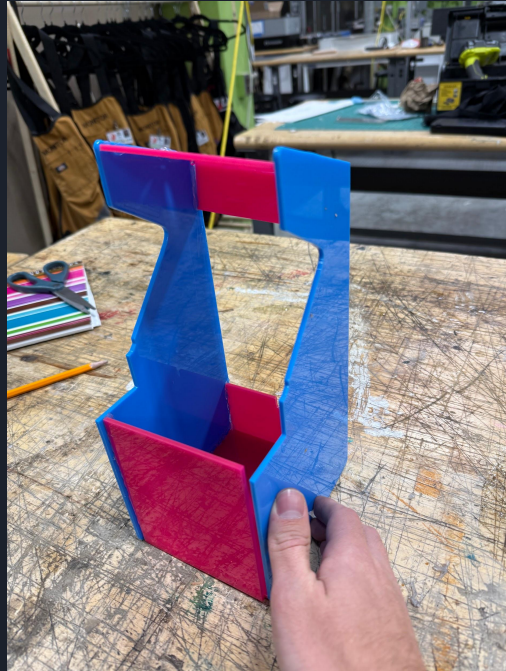
Project description

When I first arrived at Brown, there was a student made arcade cabinet in the basement of the student center. I was always a huge fan of retro arcade games when I was younger - particularly Centipede, Dig Dug, and Marvel vs. Capcom - so, I was really fond of this student creation. However, within six months, the cabinet's screen stopped working and it was thrown away. I always thought that it would be a really cool project to make a new arcade cabinet and document my process so that it could continue to be a fun activity for students for many years to come. I think this would be a good way to apply my skills in CAD, laser cutting, and design. I also think it would be a great opportunity to learn more about soldering, CNC woodworking, wood finishing, and latex printing for decals.

For my final product, I hope to have an arcade cabinet utilizing Raspberry Pi to run emulators on RetroPi. My cabinet will have 4 player capabilities, a spatial sound system, a high refresh rate display, wheels for easy transportation, and a LED marquee.

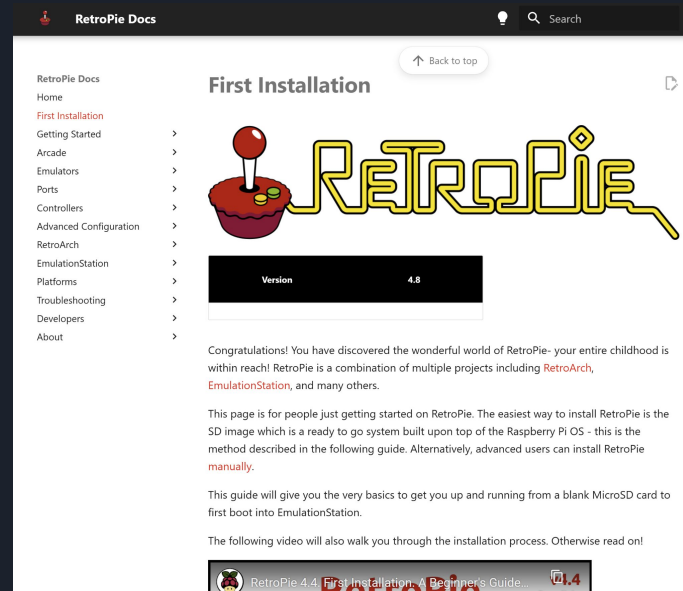
Week 1 - 9/29-10/3

Held discussion with Eiden Spilker, initial prototyping on laser cutter and designs for decals



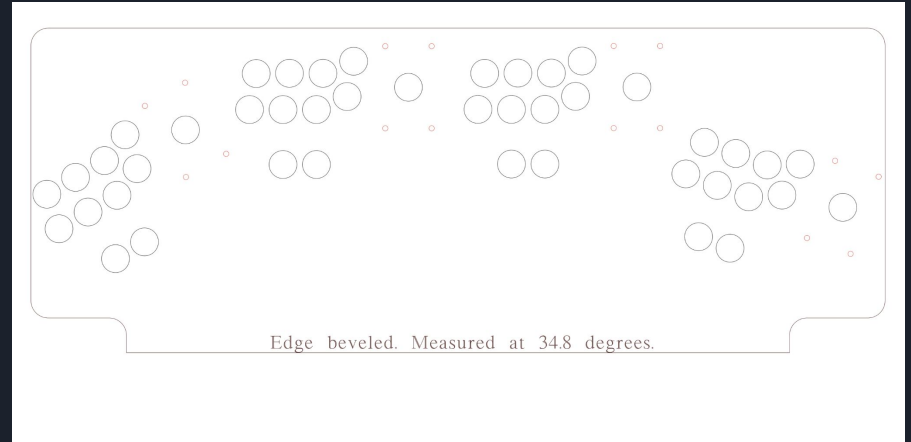
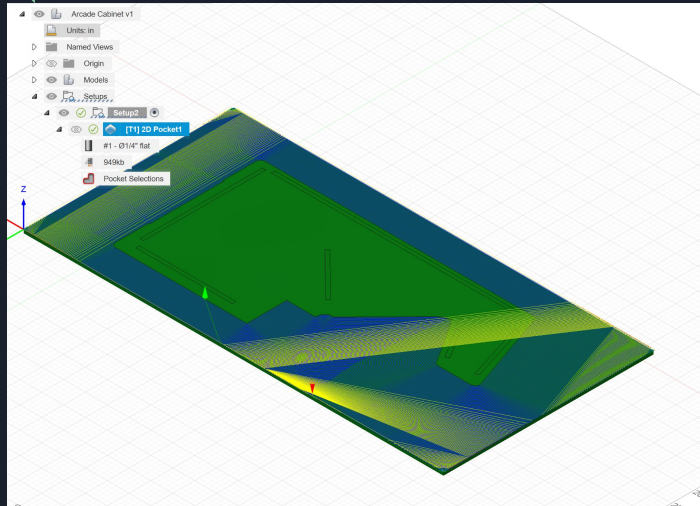
Week 2 - 10/6-10/10

Began tutorials for Raspberry Pi, completed vinyl cutter workshop



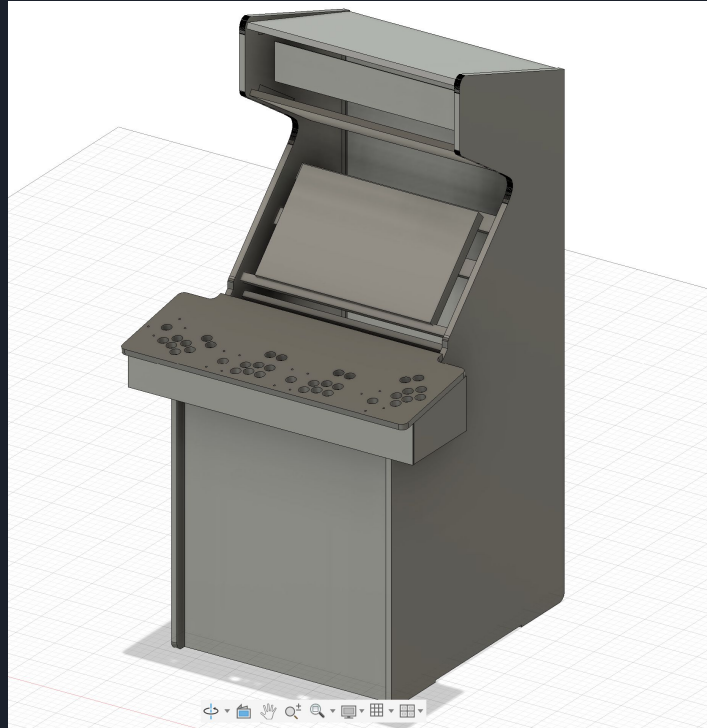
Week 3- 10/13-10/17

Modified Button Layout, Started Working on CNC Manufacturing Process



Week 4 - 10/20-10/24

Completed CAD + CAM to verify dimensions + manufacturing process



Week 4 - 10/27-10/31

Made Electronics Order for Cabinet, Completed Looks Like Prototype



EG Starts 4 Player Classic DIY Arcade Joystick Kit Parts USB Encoder To PC Controls Games + 4/8 Way Stick + 5V led Illuminated Push Buttons Compatible Video Game Consoles Mame Raspberry Pi & 4 Colors

Return or replace items: Eligible through January 31, 2026

 Buy it again



ASHGOOB 2" Caster Wheels Set of 4, Heavy Duty Casters with Brake, No Noise Locking Polyurethane (PU) Wheels, Swivel Plate Castors Pack 4

Return or replace items: Eligible through January 31, 2026

 Buy it again



Amazon Basics Stereo 2.0 Speakers for PC or Laptop, 3.5mm Aux input, USB-Powered, 1 Pair, Black

Return or replace items: Eligible through January 31, 2026

 Buy it again



CanaKit Raspberry Pi 5 Starter Kit PRO - Turbine Black (128GB Edition) (4GB RAM)

Return or replace items: Eligible through January 31, 2026

 Buy it again



Koala Holographic Sticker Paper for Inkjet Printer - 20 Sheets 8.5x11 Inch, Rainbow Printable Vinyl Sticker Paper, Waterproof, Vivid Color, Tear Resistant

 Buy it again



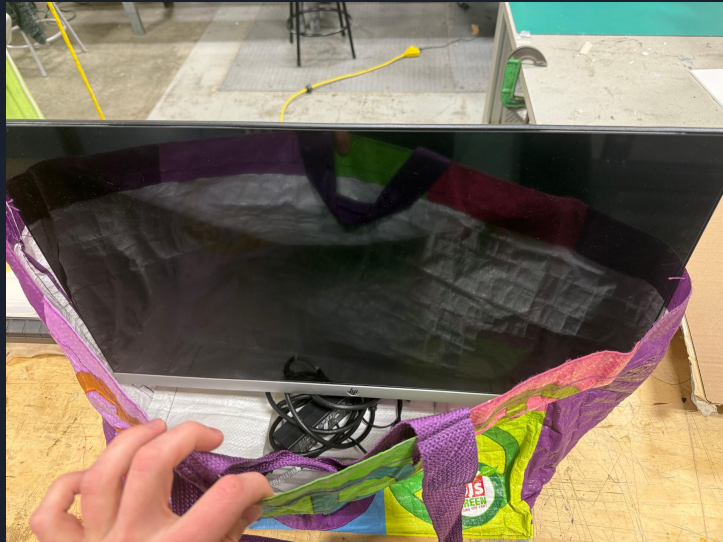
Week of 11/3-11/7

Ordered Necessary Wood Stock, Completed Electronics Assembly



Week of 11/10-11/14

Monitor and Wood Received!



Final Orders Placed, first cuts on CNC machine fully planned



Ravinte 4 Pack Soft Close Hidden H...

Delivered December 1



Rust-Oleum 1974502 Painter's Tou...

Delivered December 1



Gorilla All Purpose Wood Filler, Nat...

Delivered December 1



SENZER CS20 2 in 1 Bluetooth Com...

Delivered December 1



Atolla 4-Port USB 3.0 Hub with 4 D...

Delivered December 1



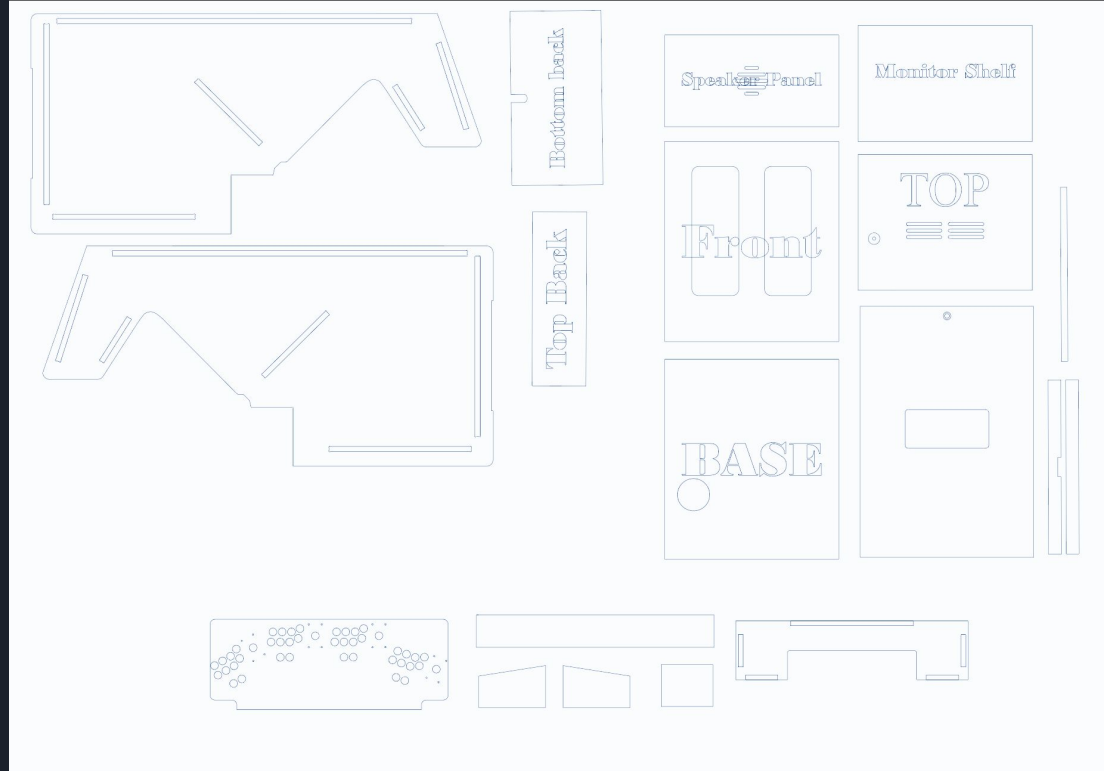
Koala Holographic Sticker Paper fo...

Delivered December 1



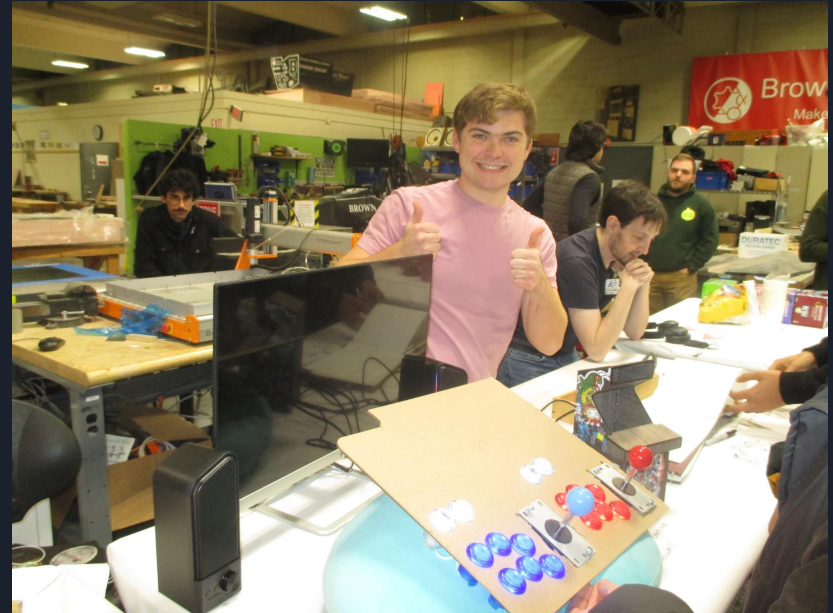
Sachie 5800 Lux High-Brightness M...

Delivered December 2



Week of 12/8

Showcase at BDW! Continued testing new components and learning woodworking techniques



January - February

Cuts completed on CNC shopbot!



January - February

Layout planned for arcade assembly + speaker placed in designated cut



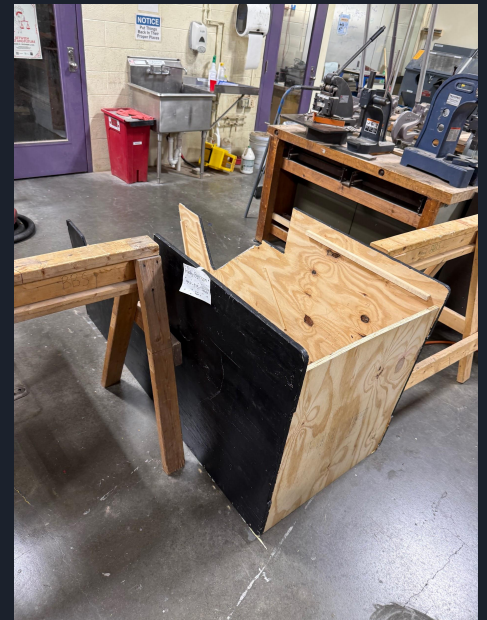
February - March

Priming + Painting arcade panels



March

Assembly of arcade controls + Arcade panels!



April

Initial arcade cabinet assembly! Mini model for reference



April

Game system installed + working, stickers started, and commemorative plaque. Final location approved in B&H lobby



April 27th

Showcase at Brown Design Fair!



brown 2026
design
fair

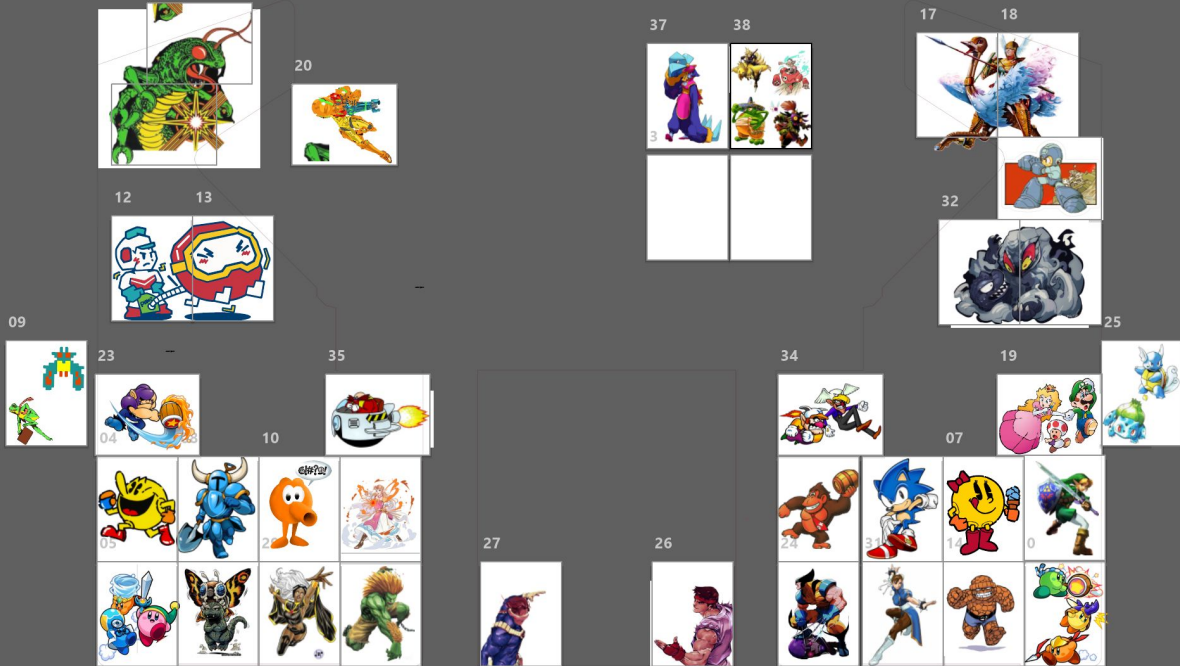
ft. free food, student art vendors, project
showcase, live music, and raffle prizes for
attendees

you want in?
scan here:



 Brown Design Workshop x Create@Brown

Remaining stickers planned + 4 Player Playtest!



May

Laser cut screen outline + LED marquee





May

All decals placed upon cabinet! Final product fully installed and running





Final reflection

This maker grant gave me the opportunity to share my love for arcade games with my fellow students at Brown. Since it has been built, I have talked to so many students who are excited to play games with their friends. I learned about CNC woodworking, Raspberry Pi OS, wood painting, and wood assembly techniques. I got to share my process with other students in the BDW and display my work alongside other makers. This grant was an incredible opportunity to do a project that I have always dreamed of and to be connected to the community at Brown. I hope the cabinet brings as much joy and inspiration to the School of Engineering as it did to me.